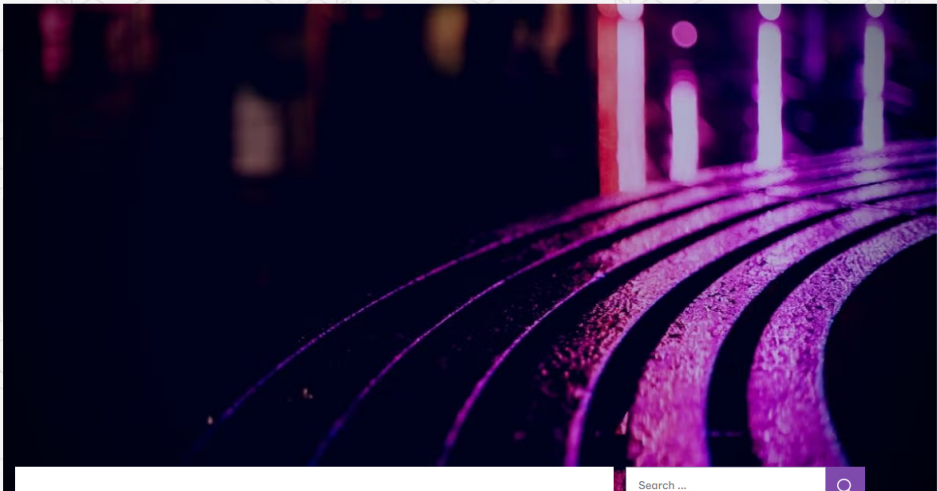




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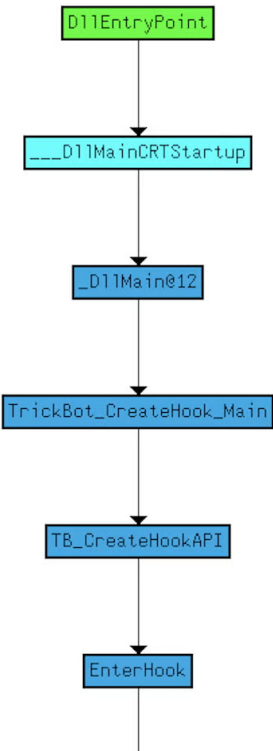
How TrickBot Malware Hooking Engine Targets Windows 10 Browsers

VITALI KREMEZ / OCTOBER 24, 2019

Vitali Kremez revealing how TrickBot's hooking engine targets Chrome, Firefox, Explorer and Edge in Windows 10.

What is TrickBot Malware? Background & Summary

TrickBot banking malware remains one of the more interesting and continually developing malware on the financial crimeware landscape. It employs multiple means and methods to exploit compromised machines of interest. The focus of this post is to cover in-depth some of its Windows 10 Microsoft Edge and other browser hooking engine functionality. We will focus on the internals, and how TrickBot leverages these browsers to set up hooks for API calls of interest. The ultimate goal of the malware browser hooking is predominantly to intercept online banking credentials before they become SSL encrypted. The stolen credentials can subsequently be used for account takeover (ATO) fraud.



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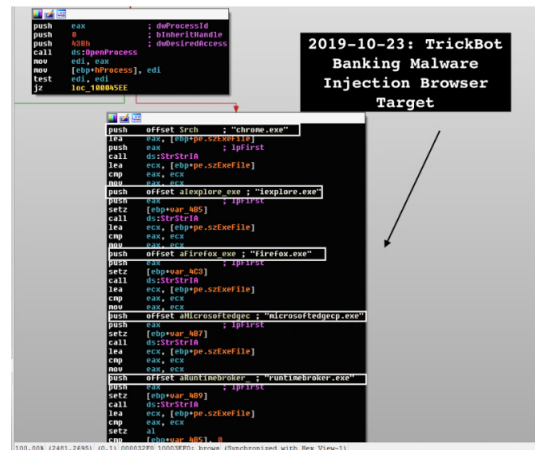
InstallHook

Since Windows 10 came with a new browser, “Microsoft Edge”, TrickBot operators needed their banking malware to operate on that software. To implement form-grabbing and web injections in the Windows 10 Edge browser, TrickBot’s rogue `rtlibroker` hooks the `microsoftedgecp.exe` process. Normally, `runtimebroker.exe` is the parent process of the Microsoft Edge browser on Windows 10 machines.

TrickBot Browser Process Injection Technique “Reflective Loader”

In order to hook browser functions, TrickBot malware injects the payload into the browser of choice via the so-called “ReflectiveLoader” methodology.

The TrickBot process injection function targets four browsers from Microsoft Edge to Google Chrome and one Microsoft Edge related process.



TrickBot injects the malware targeting the following processes:

- chrome.exe
- iexplore.exe
- firefox.exe
- microsoftedgecp.exe
- runtimebroker.exe

The malware also “relaxes” browser security and write changes files locally before injection occurs.



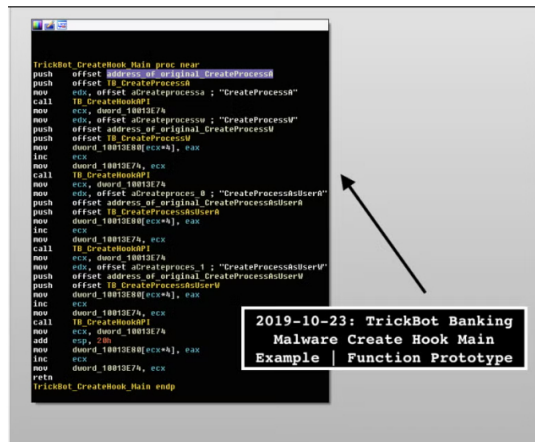
TrickBot’s reflective injection works as follows:

- Open target process and allocate memory address in remote process via `VirtualAllocEx`
- Copy function `WriteProcessMemory` into the allocated memory space
- Copy shellcode `WriteProcessMemory` into the allocated memory space
- Call `FlushInstructionCache` API to make sure our changes are written right away

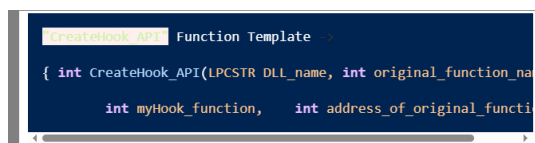
- Call inject `RemoteThread` function call
- Call `ResumeThread`
- Else, call undocumented API function `RtlCreateUserThread` to start execution in the remote process, using the offset address of the reflective loader function as the entry point.

TrickBot Malware Hooking Engine

When the TrickBot banker hooks the API function, it enters the new hooked one and checks to make sure the process is `microsoftedge.exe` while passing control to the original one when the hooked function concludes.

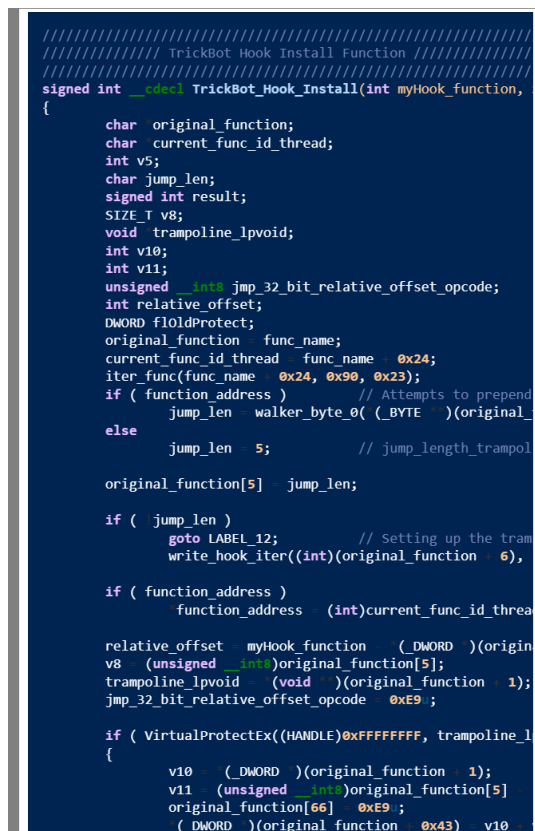


The basic TrickBot banking API hooking template is as follows:



By and large, TrickBot hooking engine works via overwriting the basic API with the redirect functions with the `Ox9` opcode, which is the call for a jump with 32-bit relative offset. TrickBot uses a trampoline function and the write hook call with the `VirtualProtectEx` API to make sure that the function has the `0x40` (`PAGE_EXECUTE_READWRITE`) property. Additionally, it attempts to conceal detection of this hooking technique via prepending NOP and/or `RETN`.

The exact TrickBot hook pseudo-code is as follows:



```
write_hook_iter(v10, jmp_32_bit_relative_offs
VirtualProtectEx( // Return to o
(HANDLE)0xFFFFFFFF,
(LPVOID )(original_function + 1),
(unsigned __int64)original_function[5],
floodProtect,
floodProtect);

result = 1;
```

For instance, TrickBot malware sets up its own custom `myCreateProcessA` function prototype after the hook on `CreateProcessA`. The idea is to catch any instance of `microsoftedge.exe` execution to intercept it for subsequent injection. This function ultimately returns the flow back to `CreateProcessA` after intercepting and collecting necessary process execution information.

```
.text:10001000 mov     ecx, offset unk_10010100
.text:10001001 push    ecx
.text:10001002 inc     ecx
.text:10001003 lock xadd [ecx], ecx
.text:10001004 push    ecx
.text:10001005 push    ecx
.text:10001006 call    @HijackForSingleInject
.text:10001007 push    ecx
.text:10001008 call    @HijackForSingleInject
.text:10001009 mov     esi, [ebp+ipfirst]
.text:1000100A test    esi, esi
.text:1000100B mov     esi, [ebp+arg_4]
.text:1000100C inc     ecx
.text:1000100D test    esi, esi
.text:1000100E test    esi, esi
.text:1000100F push    offset str_1000100F
.text:10001010 call    @HijackForSingleInject
.text:10001011 call    @HijackForSingleInject
.text:10001012 test    eax, eax
.text:10001013 jnz     short loc_10001015
.text:10001014
.text:10001015 loc_10001015:
.text:10001016 test    esi, esi
.text:10001017 jf      short loc_10001019
.text:10001018 push    offset str_10001018
.text:10001019 call    @HijackForSingleInject
.text:1000101A test    eax, eax
.text:1000101B jnz     short loc_10001015
.text:1000101C
.text:1000101D loc_1000101D:
.text:1000101E test    esi, esi
.text:1000101F jf      short loc_10001021
.text:10001020 push    offset str_10001020
.text:10001021 call    @HijackForSingleInject
.text:10001022 test    eax, eax
.text:10001023 jnz     short loc_10001025
.text:10001024
.text:10001025 loc_10001025:
.text:10001026 push    [ebp+arg_2]
.text:10001027 mov     eax, [ebp+arg_1]
.text:10001028 push    [ebp+arg_2]
.text:10001029 and     eax, 0FFFFFFFh
.text:1000102A push    [ebp+arg_1]
.text:1000102B push    [ebp+arg_2]
.text:1000102C push    [ebp+arg_1]
.text:1000102D push    [ebp+arg_2]
.text:1000102E push    [ebp+arg_1]
.text:1000102F push    [ebp+arg_2]
.text:10001030 push    [ebp+arg_1]
.text:10001031 push    [ebp+arg_2]
.text:10001032 push    [ebp+arg_1]
.text:10001033 push    [ebp+arg_2]
.text:10001034 push    [ebp+arg_1]
.text:10001035 push    [ebp+arg_2]
.text:10001036 push    [ebp+arg_1]
.text:10001037 push    [ebp+arg_2]
.text:10001038 push    [ebp+arg_1]
.text:10001039 push    [ebp+arg_2]
.text:1000103A call    address of original CreateProcessA
.text:1000103B mov     esi, eax
.text:1000103C call    @HijackForSingleInject
```

The following four API calls being hooked are in the child Microsoft Edge via rogue `rtlbroker.dll`, allowing TrickBot operators to intercept and manipulate Microsoft Edge calls:

- CreateProcess
- CreateProcessW
- CreateProcessAsUserA
- CreateProcessAsUserW

TrickBot hooks Internet Explorer and Microsoft Edge in `wininet.dll` library API calls:

- HttpSendRequestA
- HttpSendRequestW
- HttpSendRequestExA
- HttpSendRequestExW
- InternetCloseHandle
- InternetReadFile
- InternetReadFileExA
- InternetQueryDataAvailable
- HttpQueryInfoA
- InternetWriteFile
- HttpEndRequestA
- HttpEndRequestW
- InternetQueryOptionA
- InternetQueryOptionW
- InternetSetOptionA
- InternetSetOptionW
- HttpOpenRequestA
- HttpOpenRequestW
- InternetConnectA
- InternetConnectW

The malware hooks Mozilla Firefox Browser in `nspr4.dll` library API calls:

- PR_OpenTCPSocket
- PR_Connect
- PR_Close
- PR_Write
- PR_Read

It hooks Chrome in `chrome.dll` library API calls:

- ssl_read
- ssl_write

Reference

injectDll32.dll C546D40D411D0F0BB7A1C9986878F231342CDF8B

rtlbrokerDll.dll 0785D0C5600D9C096B75CC4465BE79D456F60594

testnewinj32Dll.dll D5F98BFF5E33A86B213E05344BD402350FC5F7CD

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VITALI KREMEZ

Vitali Kremez is a strategic advisor for SentinelLabs. He specializes in researching and investigating complex cyberattacks, network intrusions, data breaches, and hacking incidents mainly emanating from the Eastern European cybercriminal ecosystem. He has earned the majority of major certifications available in information technology, information security, and digital forensics fields.



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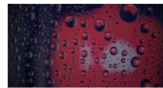
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